

What It Does:

Smart lighting system built on ESP32 that performs three coordinated functions:

1. Sensing (Input Layer)

- The LDR continuously measures ambient light intensity.
- Output is an analog signal (0–4095 via ESP32 ADC).

2. Decision Making (Processing Layer)

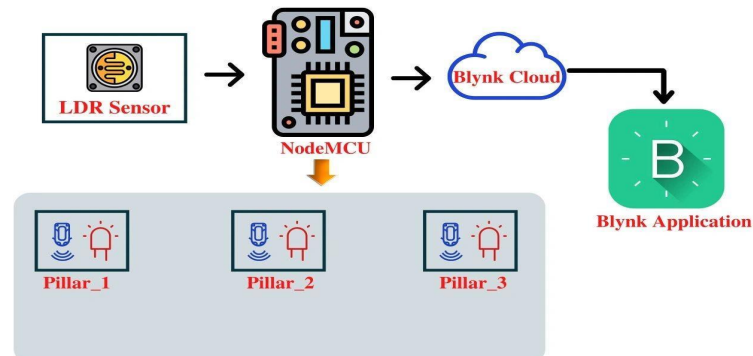
- ESP32 evaluates light level against a threshold.
- It decides whether lighting is required or not.

3. Actuation + Monitoring (Output + IoT Layer)

- LED is turned ON/OFF automatically (or manually via app).
- LCD shows real-time values locally.
- Data and control are integrated with Blynk IoT for remote interaction.

Working Explanation

Block Diagram



1. Light Detection (Sensing Stage)

- LDR resistance changes with light:
 - **Bright light** → **low resistance**
 - **Dark** → **high resistance**
- In a voltage divider:
 - This converts resistance → voltage
- ESP32 reads it via ADC:
 - Example:

- Bright → ~500–1500
- Dark → ~2500–5000

2. Signal Processing (Decision Logic)

ESP32 runs logic:

IF (LDR > threshold) → Dark → Turn ON light
 ELSE → Bright → Turn OFF light

- This is a **closed-loop reactive control system** (no human needed).

3. Actuation (Lighting Control)

- LED acts as a **load (simulated street light/home light)**
- Based on decision:
 - Dark → LED ON
 - Bright → LED OFF

4. IoT Integration (Cloud Interaction)

- ESP32 connects to WiFi and communicates with Blynk:
 - Sends:
 - LDR value (V0)
 - LED status (V2)
 - Receives:
 - Manual control command (V1)

This creates a **bidirectional control system**:

- Sensor → App (monitoring)
- App → Device (control)

5. User Interface (Dual Feedback)

- **LCD (Local Monitoring)**
 - Shows:
 - LDR value
 - LED status
- **Blynk App (Remote Monitoring)**
 - Gauge → Light level
 - Switch → Control LED
 - Indicator → LED status

Reason of Doing:

1. Energy Efficiency

- Lights operate **only when needed**
- Reduces electricity consumption

Real-world use:

- Street lighting
- Garden lights
- Corridor lighting

2. Automation

- Eliminates manual switching
- Works continuously without human intervention

3. Remote Control Capability

- Using Blynk IoT:
 - You can control lighting from anywhere
 - Override automatic behavior

4. Real-Time Monitoring

- Continuous feedback:
 - Sensor value
 - Device status

Important for:

- Smart homes
- Industrial automation

5. Scalable IoT System

- This is a **basic node of a smart grid**
- Can be expanded to:
 - Multiple lights
 - City-wide automation